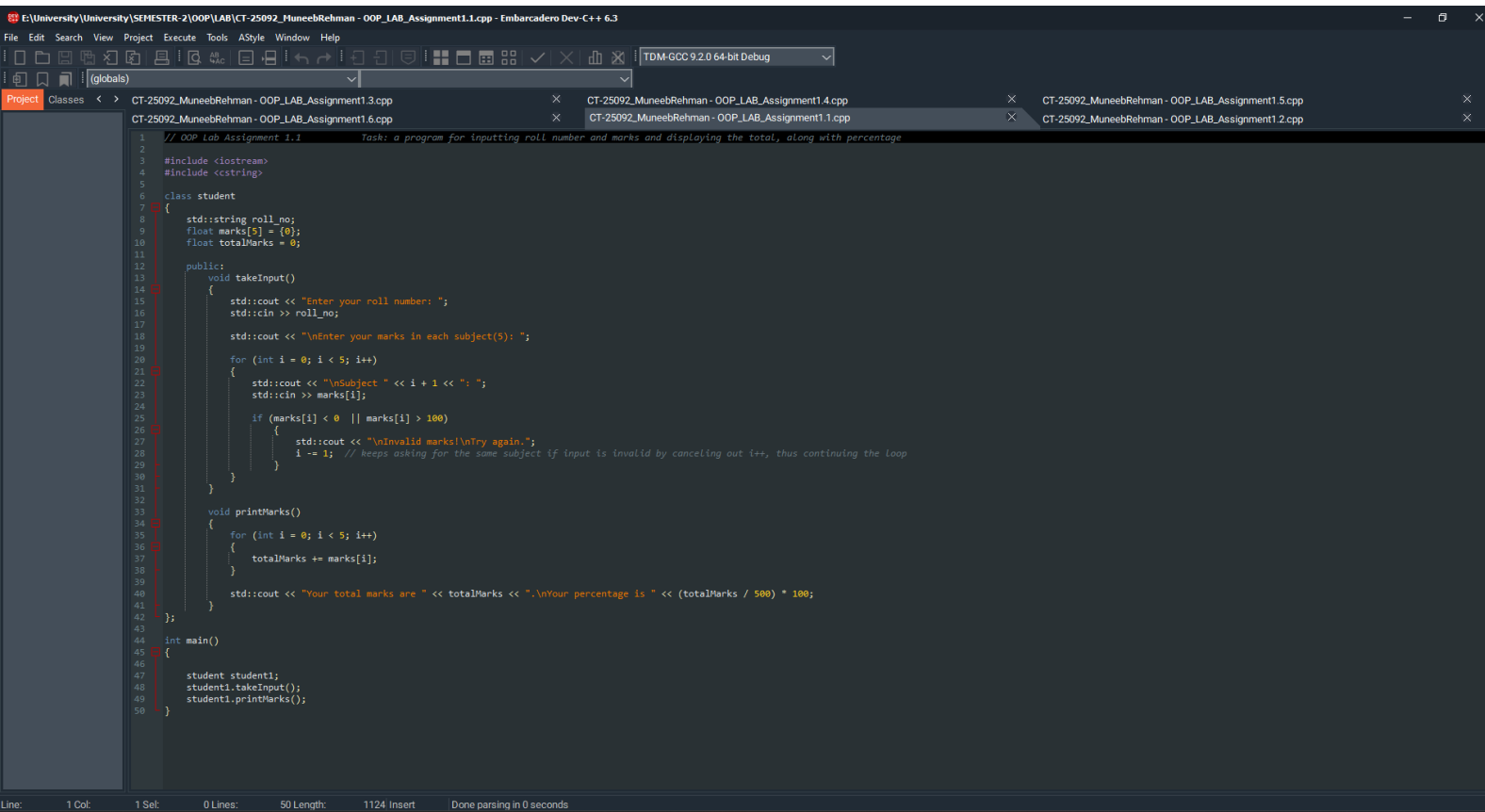


Question 1.



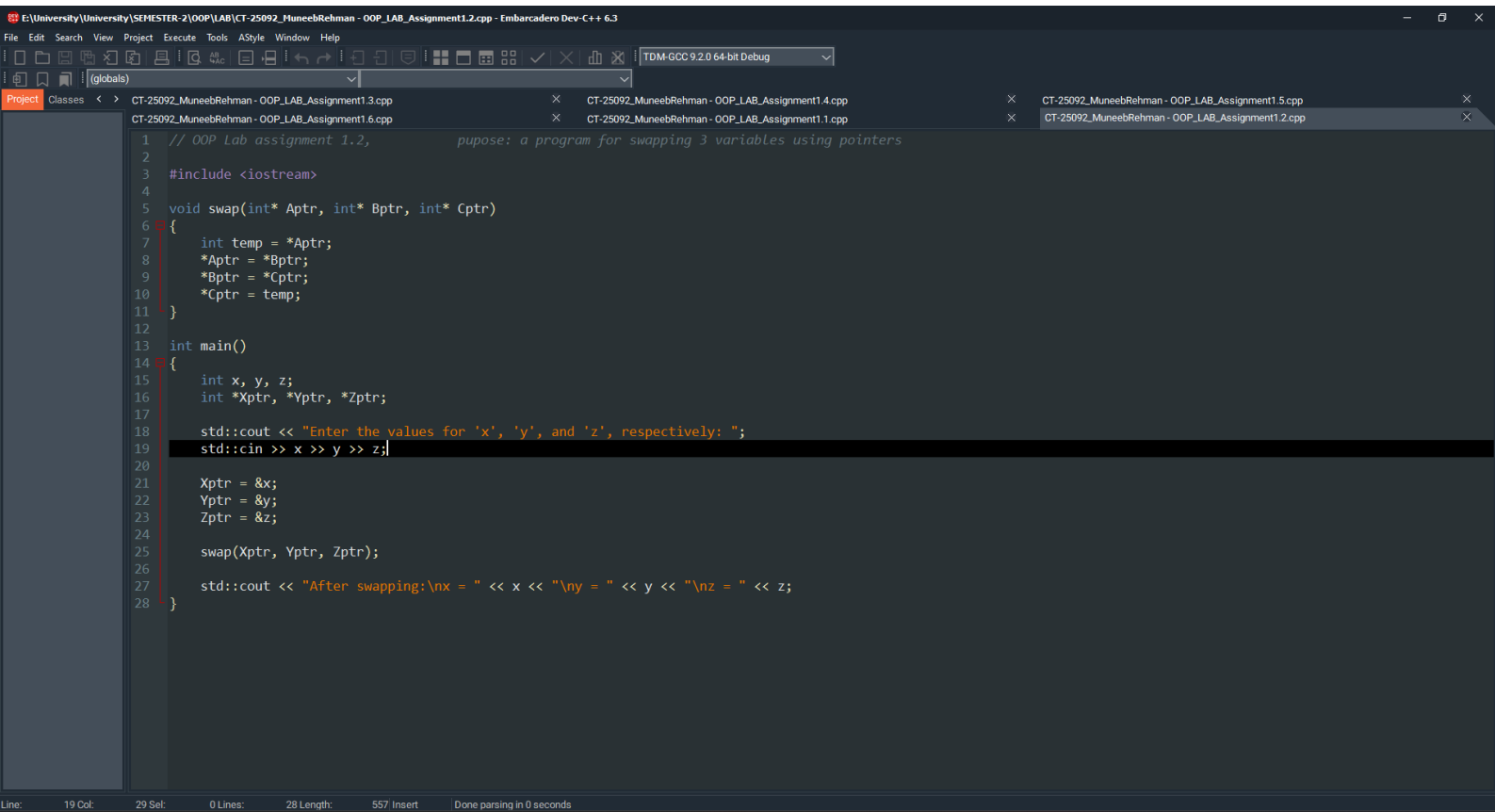
The screenshot shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment1.1.cpp". The code is for a program that takes roll numbers and marks for 5 subjects, calculates the total marks, and displays the percentage.

```
1 // OOP Lab Assignment 1.1 Task: a program for inputting roll number and marks and displaying the total, along with percentage
2
3 #include <iostream>
4 #include <string>
5
6 class student
7 {
8     std::string roll_no;
9     float marks[5] = {0};
10    float totalMarks = 0;
11
12 public:
13     void takeInput()
14     {
15         std::cout << "Enter your roll number: ";
16         std::cin >> roll_no;
17
18         std::cout << "\nEnter your marks in each subject(5): ";
19
20         for (int i = 0; i < 5; i++)
21         {
22             std::cout << "\nSubject " << i + 1 << ": ";
23             std::cin >> marks[i];
24
25             if (marks[i] < 0 || marks[i] > 100)
26             {
27                 std::cout << "\nInvalid marks!\nTry again.";
28                 i -= 1; // keeps asking for the same subject if input is invalid by canceling out i++, thus continuing the loop
29             }
30         }
31
32     }
33
34     void printMarks()
35     {
36         for (int i = 0; i < 5; i++)
37         {
38             totalMarks += marks[i];
39         }
40
41         std::cout << "Your total marks are " << totalMarks << ".\nYour percentage is " << (totalMarks / 500) * 100;
42     }
43 };
44
45 int main()
46 {
47     student student1;
48     student1.takeInput();
49     student1.printMarks();
50 }
```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, and a project explorer on the left. The status bar at the bottom shows "Line: 1 Col: 1 Sel: 0 Lines: 50 Length: 1124 Insert Done parsing in 0 seconds".

```
E:\University\University\SEME x + v
Enter your roll number: CT-092
Enter your marks in each subject(5):
Subject 1: 79
Subject 2: 78
Subject 3: 82
Subject 4: 80
Subject 5: 81
Your total marks are 400.
Your percentage is 80
-----
Process exited after 17.22 seconds with return value 0
Press any key to continue . . . |
```

Question 2.



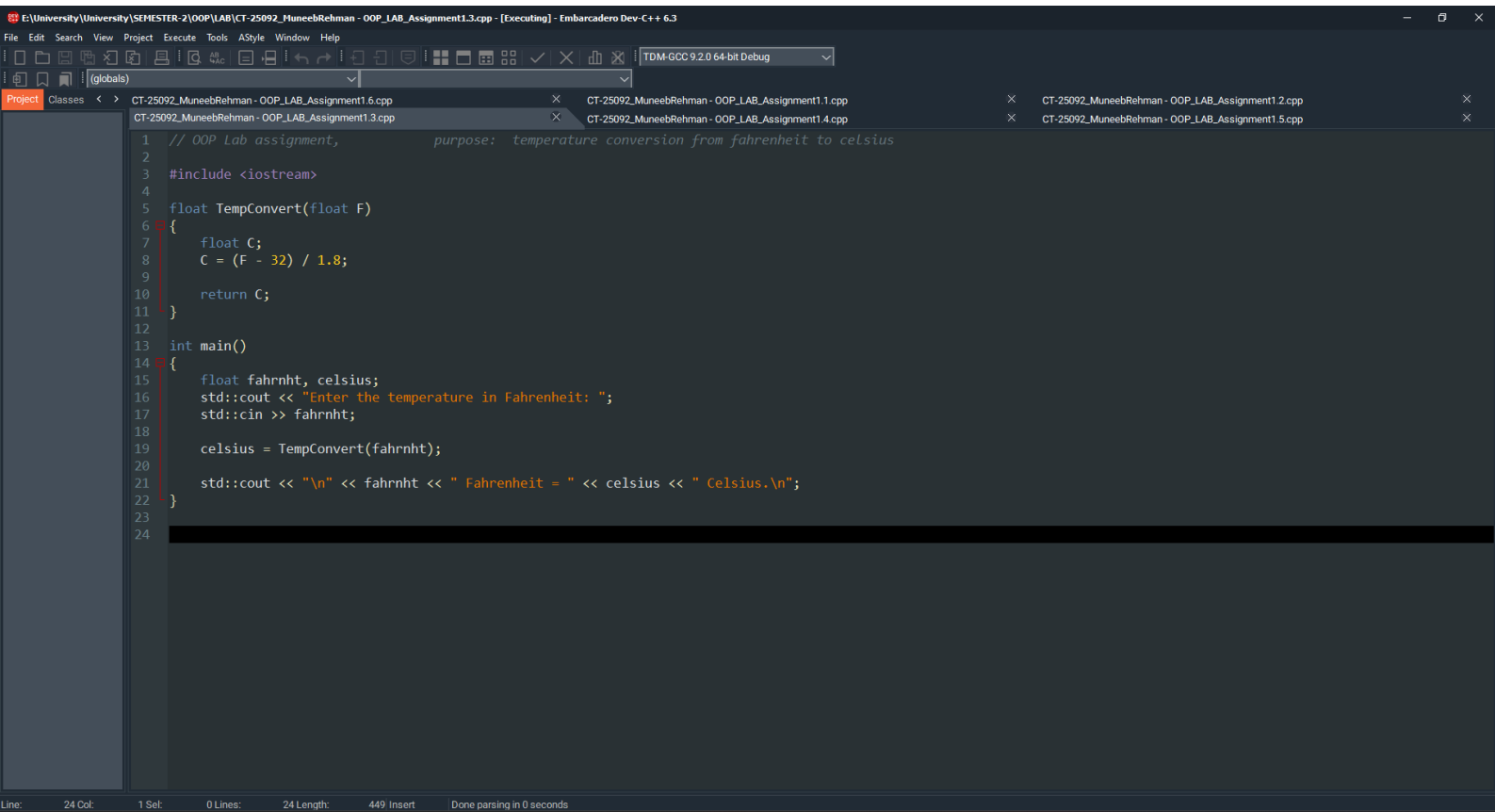
The screenshot shows a code editor window with the following C++ code:

```
1 // OOP Lab assignment 1.2, pupose: a program for swapping 3 variables using pointers
2
3 #include <iostream>
4
5 void swap(int* Aptr, int* Bptr, int* Cptr)
6 {
7     int temp = *Aptr;
8     *Aptr = *Bptr;
9     *Bptr = *Cptr;
10    *Cptr = temp;
11 }
12
13 int main()
14 {
15     int x, y, z;
16     int *Xptr, *Yptr, *Zptr;
17
18     std::cout << "Enter the values for 'x', 'y', and 'z', respectively: ";
19     std::cin >> x >> y >> z;
20
21     Xptr = &x;
22     Yptr = &y;
23     Zptr = &z;
24
25     swap(Xptr, Yptr, Zptr);
26
27     std::cout << "After swapping:\nx = " << x << "\ny = " << y << "\nz = " << z;
28 }
```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, and a project explorer on the left. The status bar at the bottom displays: Line: 19 Col: 29 Sel: 0 Lines: 28 Length: 557 Insert Done parsing in 0 seconds.

```
E:\University\University\SEME x + v
Enter the values for 'x', 'y', and 'z', respectively: 12
15
19
After swapping:
x = 15
y = 19
z = 12
=====
Process exited after 12.65 seconds with return value 0
Press any key to continue . . . |
```

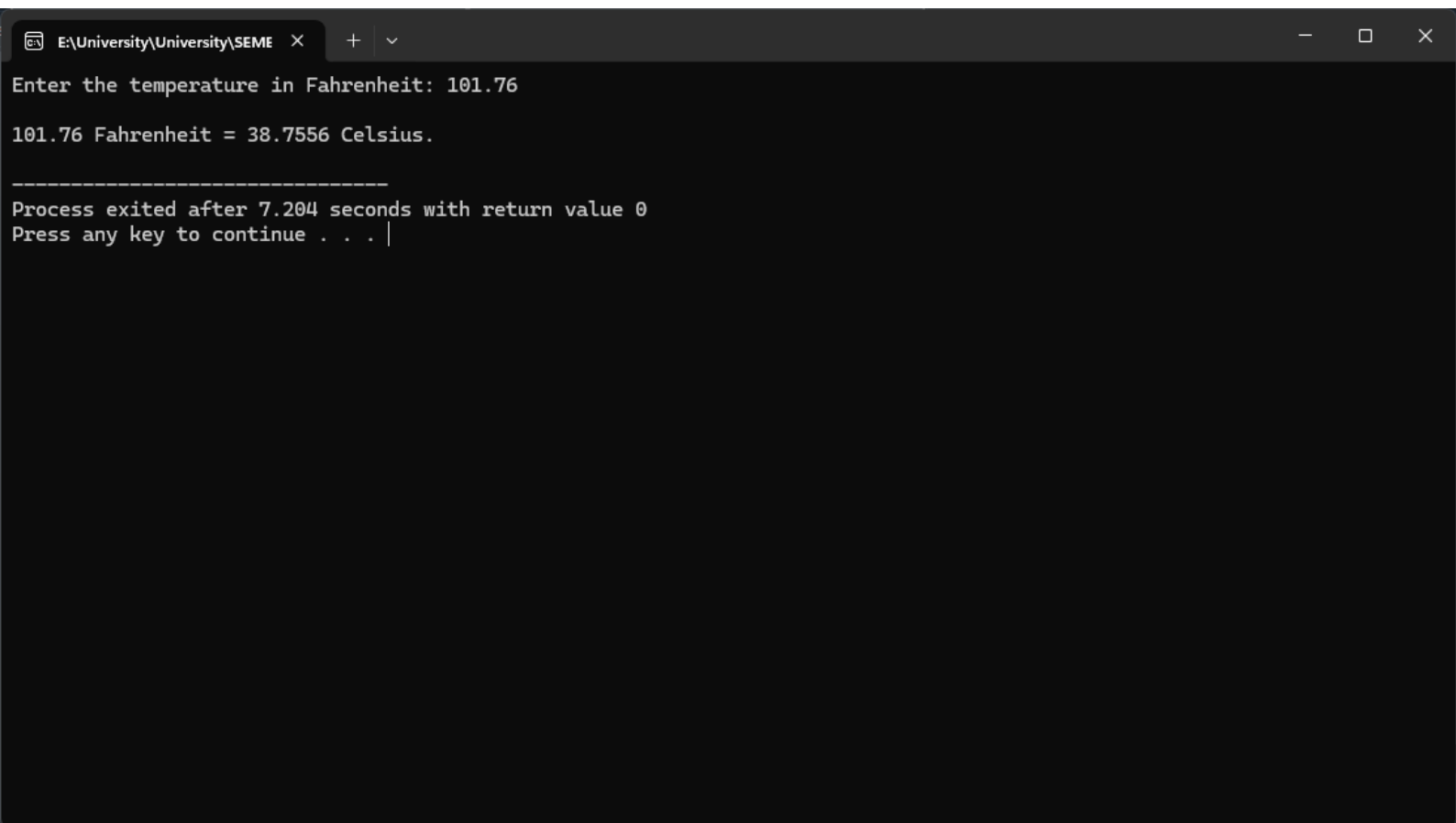
Question 3



The screenshot shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp". The code is for a temperature conversion program from Fahrenheit to Celsius. The program includes `<iostream>` and defines a function `TempConvert` that takes a float `F` and returns a float `C`. The `main` function prompts the user to enter a temperature in Fahrenheit, reads the input, converts it using `TempConvert`, and displays the result in Celsius.

```
1 // OOP Lab assignment,           purpose: temperature conversion from fahrenheit to celsius
2
3 #include <iostream>
4
5 float TempConvert(float F)
6 {
7     float C;
8     C = (F - 32) / 1.8;
9
10    return C;
11 }
12
13 int main()
14 {
15     float fahrnht, celsius;
16     std::cout << "Enter the temperature in Fahrenheit: ";
17     std::cin >> fahrnht;
18
19     celsius = TempConvert(fahrnht);
20
21     std::cout << "\n" << fahrnht << " Fahrenheit = " << celsius << " Celsius.\n";
22 }
23
24
```

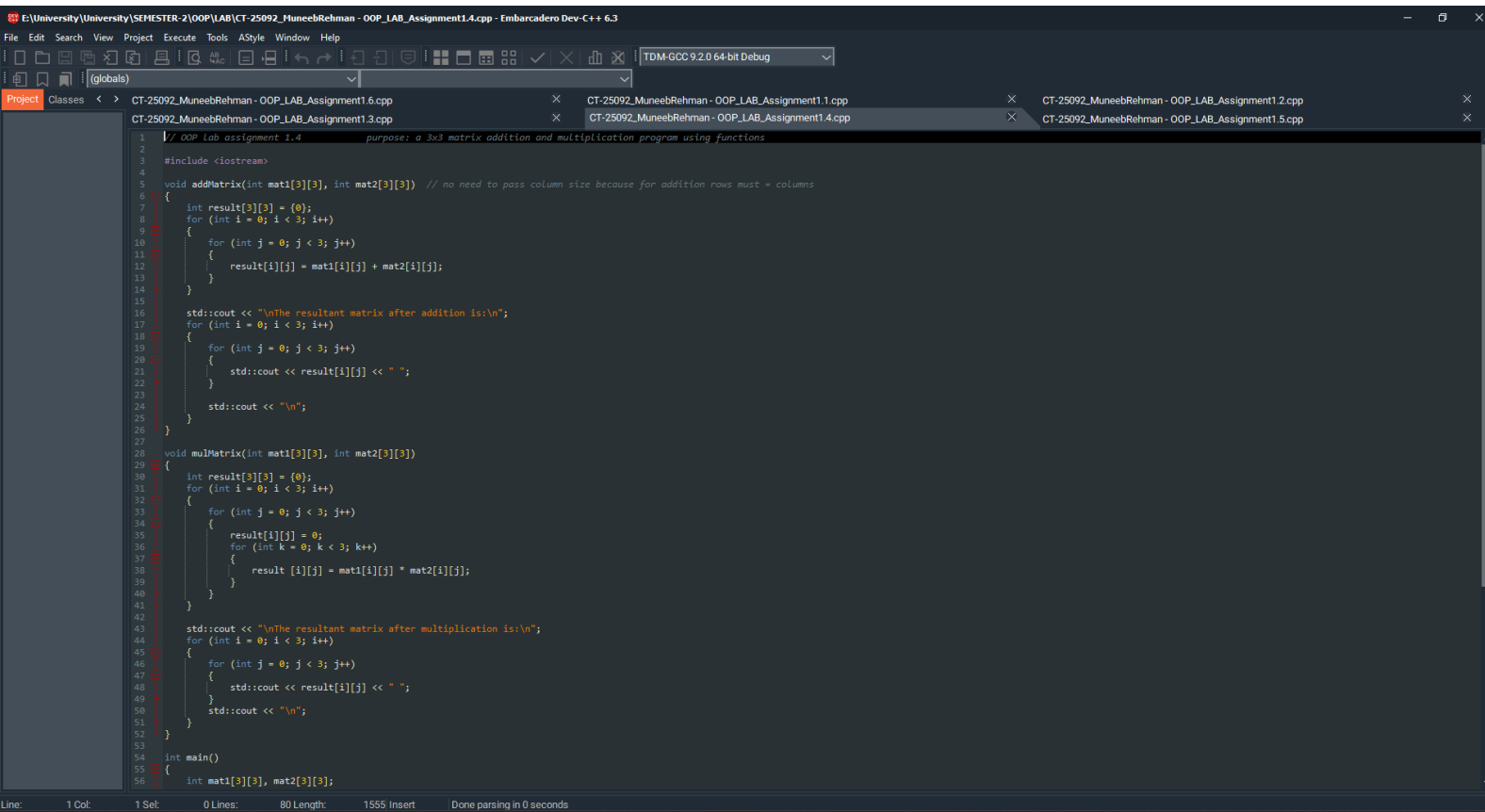
The status bar at the bottom indicates: Line: 24 Col: 1 Sel: 0 Lines: 24 Length: 449 Insert: Done parsing in 0 seconds.



A screenshot of a Windows command prompt window. The title bar shows the file path "E:\University\University\SEME" and standard window controls. The command prompt displays the following text:

```
Enter the temperature in Fahrenheit: 101.76  
101.76 Fahrenheit = 38.7556 Celsius.  
-----  
Process exited after 7.204 seconds with return value 0  
Press any key to continue . . . |
```

Question 4



The screenshot shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp". The code is for a 3x3 matrix addition and multiplication program. It includes `<iostream>` and defines two functions: `addMatrix` and `mulMatrix`. The `addMatrix` function takes two 3x3 matrices and returns their sum. The `mulMatrix` function takes two 3x3 matrices and returns their product. The `main` function calls both functions and prints the results.

```
1 // OOP Lab assignment 1.4 purpose: a 3x3 matrix addition and multiplication program using functions
2
3 #include <iostream>
4
5 void addMatrix(int mat1[3][3], int mat2[3][3]) // no need to pass column size because for addition rows must = columns
6 {
7     int result[3][3] = {0};
8     for (int i = 0; i < 3; i++)
9     {
10         for (int j = 0; j < 3; j++)
11         {
12             result[i][j] = mat1[i][j] + mat2[i][j];
13         }
14     }
15
16     std::cout << "\nThe resultant matrix after addition is:\n";
17     for (int i = 0; i < 3; i++)
18     {
19         for (int j = 0; j < 3; j++)
20         {
21             std::cout << result[i][j] << " ";
22         }
23         std::cout << "\n";
24     }
25 }
26
27 void mulMatrix(int mat1[3][3], int mat2[3][3])
28 {
29     int result[3][3] = {0};
30     for (int i = 0; i < 3; i++)
31     {
32         for (int j = 0; j < 3; j++)
33         {
34             result[i][j] = 0;
35             for (int k = 0; k < 3; k++)
36             {
37                 result[i][j] = mat1[i][k] * mat2[k][j];
38             }
39         }
40     }
41
42     std::cout << "\nThe resultant matrix after multiplication is:\n";
43     for (int i = 0; i < 3; i++)
44     {
45         for (int j = 0; j < 3; j++)
46         {
47             std::cout << result[i][j] << " ";
48         }
49         std::cout << "\n";
50     }
51 }
52
53 int main()
54 {
55     int mat1[3][3], mat2[3][3];
56 }
```

Line: 1 Col: 1 Sel: 0 Lines: 80 Length: 1555 Insert Done parsing in 0 seconds

E:\University\University\SEMESTER-2\OOP\LAB\CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp - Embarcadero Dev-C++ 6.3

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 9.2.0 64-bit Debug

Project Classes >> CT-25092_MuneebRehman - OOP_LAB_Assignment1.6.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment1.1.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment1.2.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp CT-25092_MuneebRehman - OOP_LAB_Assignment1.5.cpp

```
25 }
26 }
27
28 void mulMatrix(int mat1[3][3], int mat2[3][3])
29 {
30     int result[3][3] = {0};
31     for (int i = 0; i < 3; i++)
32     {
33         for (int j = 0; j < 3; j++)
34         {
35             result[i][j] = 0;
36             for (int k = 0; k < 3; k++)
37             {
38                 result[i][j] = mat1[i][k] * mat2[k][j];
39             }
40         }
41     }
42
43     std::cout << "\nThe resultant matrix after multiplication is:\n";
44     for (int i = 0; i < 3; i++)
45     {
46         for (int j = 0; j < 3; j++)
47         {
48             std::cout << result[i][j] << " ";
49         }
50         std::cout << "\n";
51     }
52 }
53
54 int main()
55 {
56     int mat1[3][3], mat2[3][3];
57
58     std::cout << "Enter the elements for the first matrix: ";
59     for (int i = 0; i < 3; i++)
60     {
61         for (int j = 0; j < 3; j++)
62         {
63             std::cin >> mat1[i][j];
64         }
65     }
66
67     std::cout << "\nEnter the elements for the second matrix: ";
68     for (int i = 0; i < 3; i++)
69     {
70         for (int j = 0; j < 3; j++)
71         {
72             std::cin >> mat2[i][j];
73         }
74     }
75
76     addMatrix(mat1, mat2);
77     mulMatrix(mat1, mat2);
78 }
79
80 }
```

Line: 1 Col: 1 Sel: 0 Lines: 80 Length: 1555 Insert Done parsing in 0 seconds

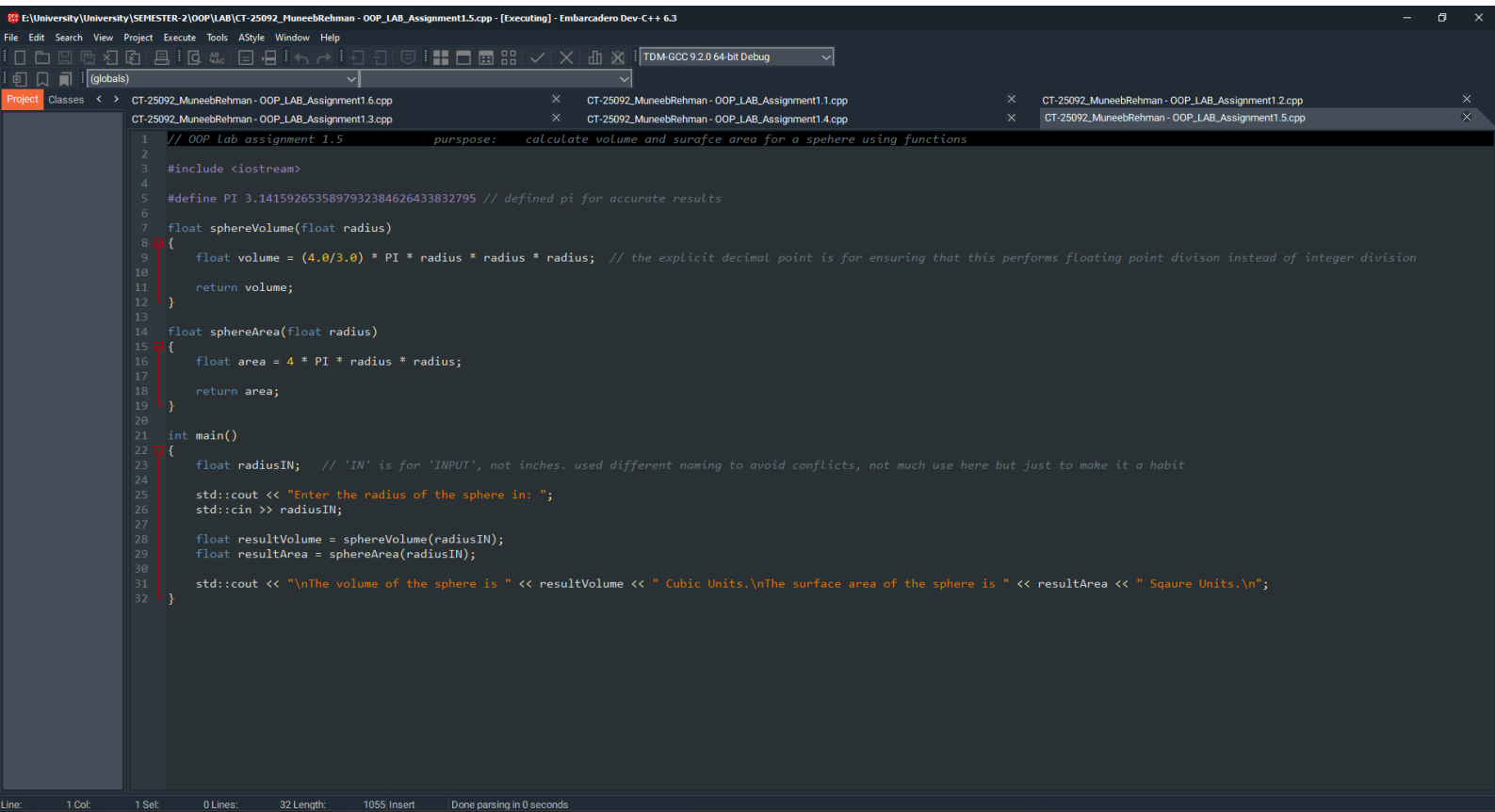

```
E:\University\University\SEME x + v
Enter the elements for the first matrix: 1 2 3 4 5 6 7 8 9
Enter the elements for the second matrix: 2 4 6 8 10 12 14 16 18

The resultant matrix after addition is:
3 6 9
12 15 18
21 24 27

The resultant matrix after multiplication is:
2 8 18
32 50 72
98 128 162

-----
Process exited after 29.99 seconds with return value 0
Press any key to continue . . . |
```

Question 5.



The screenshot shows a C++ IDE with a project named "CT-25092_MuneebRehman - OOP_LAB_Assignment1.5.cpp". The code is for a program that calculates the volume and surface area of a sphere using functions. The code is as follows:

```
1 // OOP lab assignment 1.5 purpose: calculate volume and surafce area for a spehere using functions
2
3 #include <iostream>
4
5 #define PI 3.1415926535897932384626433832795 // defined pi for accurate results
6
7 float sphereVolume(float radius)
8 {
9     float volume = (4.0/3.0) * PI * radius * radius * radius; // the explicit decimal point is for ensuring that this performs floating point divison instead of integer division
10    return volume;
11 }
12
13
14 float sphereArea(float radius)
15 {
16     float area = 4 * PI * radius * radius;
17
18     return area;
19 }
20
21 int main()
22 {
23     float radiusIN; // 'IN' is for 'INPUT', not inches. used different naming to avoid conflicts, not much use here but just to make it a habit
24
25     std::cout << "Enter the radius of the sphere in: ";
26     std::cin >> radiusIN;
27
28     float resultVolume = sphereVolume(radiusIN);
29     float resultArea = sphereArea(radiusIN);
30
31     std::cout << "\nThe volume of the sphere is " << resultVolume << " Cubic Units.\nThe surface area of the sphere is " << resultArea << " Sqaure Units.\n";
32 }
```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, and a project explorer on the left. The status bar at the bottom shows "Line: 1 Col: 1 Sel: 0 Lines: 32 Length: 1055 Insert Done parsing in 0 seconds".

```
E:\University\University\SEME x + v
Enter the radius of the sphere in: 5.789

The volume of the sphere is 812.642 Cubic Units.
The surface area of the sphere is 421.131 Sqaure Units.

-----
Process exited after 6.377 seconds with return value 0
Press any key to continue . . . |
```

Question 6.

```
1 // OOP assihnment 1.6           purpose:   Bank cash withdrawal system
2
3 #include <iostream>
4 #include <string>
5
6 class bankAccount
7 {
8     private:
9         std::string acc_no; // account no
10        int acc_pin; // account pin
11        char acc_type;
12        float balance; // account balance
13
14    public:
15        bankAccount(std::string accNo, int accPin, char accType)
16        {
17            acc_no = accNo;
18            acc_pin = accPin;
19            acc_type = accType;
20            balance = 200000; // default balance of 200k
21        }
22
23        void withdraw(float amount)
24        {
25            if (amount > 100000)
26            {
27                std::cout << "\nAmount cannot exceed 100,000!";
28                return;
29            }
30
31            if (acc_type == 'S' || acc_type == 's')
32            {
33                if (amount <= 50000)
34                {
35                    balance -= amount - (amount * 0.02); // 2% transaction fee
36                }
37
38                else if (amount > 50000)
39                {
40                    balance -= amount - (amount * 0.05); // 5% transaction fee for withdrawal above 50k
41                }
42            }
43
44            else if (acc_type == 'C' || acc_type == 'c')
45            {
46                if (amount <= 50000)
47                {
48                    balance -= amount - 100; // 100 transaction fee
49                }
50
51                else if (amount > 50000)
52                {
53                    balance -= amount - (amount * 0.05); // 5% transaction fee for withdrawal above 50k
54                }
55            }
56        }
57
58        std::cout << "\nTransaction successful!\nPlease, take your card and cash.";
59        std::cout << "\nYour new balance is " << balance;
60
61    }
62 }
```

```

6     }
7
8     std::cout << "\nTransaction successful!\nPlease, take your card and cash.";
9     std::cout << "\nYour new balance is " << balance;
10
11 }
12
13 };
14
15 int main()
16 {
17     std::string accountNo;
18     std::cout << "Enter you account number: ";
19     std::cin >> accountNo;
20
21     int accountPin;
22     std::cout << "\nEnter your pin: ";
23     std::cin >> accountPin;
24
25     char accountType;
26     std::cout << "\nEnter your account type: ";
27     std::cin >> accountType;
28
29     bankAccount user1(accountNo, accountPin, accountType);
30
31     float userAmount;
32     std::cout << "\nEnter the amount you wish to withdraw: ";
33     std::cin >> userAmount;
34
35     user1.withdraw(userAmount);
36 }

```

```
E:\University\University\SEME x + v
Enter you account number: IBN-67341
Enter your pin: 4538
Enter your account type: s
Enter the amount you wish to withdraw: 45900
Transaction successful!
Please, take your card and cash.
Your new balance is 155018
-----
Process exited after 27.43 seconds with return value 0
Press any key to continue . . . |
```